Track 1 · Smart Campus: forecasting hourly building energy with an extrapolating linear model and learned feature imputation

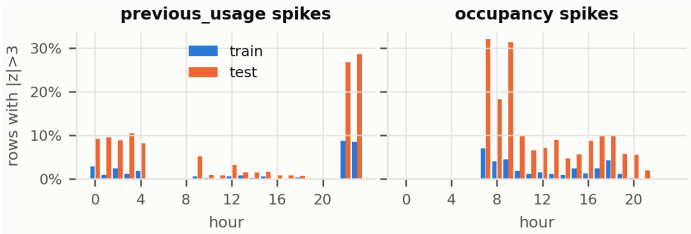
Team Cookie Squad · Final model c04 · Public RMSE **3.33681** (c03 3.36875) · CV RMSE 3.048 clean / 3.228 test-like · R^2 0.973

1 · Problem

Predict *energy_usage* for 12 buildings (8 types) from building, hour / weekday / month, temperature, humidity, occupancy and *previous_usage*; metric RMSE. Test rows arrive shuffled without ids, so all features are row-local. *previous_usage* is not the previous hour: its daily profile matches energy at the **same** hour (residential corr. 0.97 vs 0.52 one hour back) and rows never chain, so it acts as a same-slot reference from an earlier period.

2 · Critical insight: test is shifted, and the extremes are real

Test has 3× train's missing rate (18.2% vs 5.8% of rows) and, using a target-free consistency model (each feature predicted from the others), **16.6%** spike rows (|z|>3) vs **4.1%** in train. The spikes follow a schedule (prev 22:00–04:00, occupancy 07:00–09:00). They are not errors: on train spike rows the energy left unexplained without the feature tracks its deviation (r = 0.92) along the same line as ordinary rows. So the model must **extrapolate**, and trees, which saturate beyond the training range, are the wrong tool for the energy equation.



3 · Validation design

Paired 5-fold × 4-repeat CV, scored three ways: **clean**; **test-like** (validation rows receive missing-value masks sampled from the test file's joint missing patterns); **shift-weighted** (test-like, with spike rows up-weighted ×4.05 to test frequency). A train-vs-test classifier on features (AUC 0.70) independently gave spike rows ≈3× weight. Imputers never see the target, and validation rows are excluded from imputer fitting inside each fold.

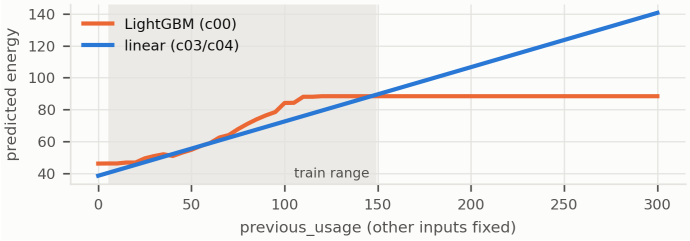
4 · Method

Energy model (linear ridge, α=3): per-building intercept, occupancy and temperature slopes; shared *previous_usage* (≈0.34) and humidity slopes; smooth daily profile per building type (Fourier) with ridge-shrunk hour corrections; weekend shape; annual season; temperature². **Missing inputs:** c03 used a reduced sub-model per missing pattern. Error analysis showed this distorts correlated slopes: without occupancy, the *previous_usage* slope inflates from 0.34 to 0.46 and mis-extrapolates on prev spikes. **c04** instead fills each missing input with a LightGBM imputer (features only; train + unlabelled test features; test-like mask augmentation) and always applies the **full** linear model. LightGBM estimates inputs; the energy prediction stays linear.

5 · Results

RMSE	Clean	Test-like	Shift-wtd	Spike	Public
c00 LightGBM	3.341	3.964	4.281	5.967	–
c03 linear	3.077	3.296	3.543	4.875	3.36875
c04 final	**3.048**	**3.228**	**3.363**	**4.150**	**3.33681**

c04 beats c03 in all 4/4 repeats on every view (test-like Δ -0.068, shift-weighted -0.180, non-spike -0.029; test-like MSE Δ -0.45, 95% CI [-0.67, -0.25]). The public score confirmed the direction (3.36875 → 3.33681); the smaller size is within public-set noise.

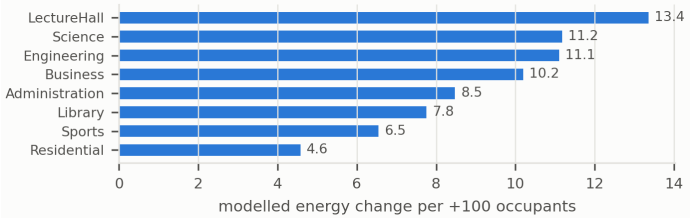


6 · Robustness and error analysis

Our validation indicates the remaining error is noise: LightGBM, ExtraTrees, CatBoost and an MLP trained on c04's out-of-fold residuals all reach out-of-sample R^2 below 0 (–0.02 to –0.004), and the best correction weight is zero. Group-specific slopes, a separate slope for spiked values, density-ratio weighting, other imputers and blends did not improve. Forensics found no exploitable sequence: *previous_usage* matches the next hour's energy at chance rate (0.5% vs 0.3–0.7% controls) and ids are randomly ordered. Imputers fitted with test features are no worse than train-only imputers on held-out train-like rows and clearly better on held-out test-like rows.

7 · Campus insights (associations, not causal effects)

+100 occupants ≈ **+13.4** energy units in the lecture hall vs **+4.6** in residences, so lecture halls are the most occupancy-driven. Science buildings carry the largest modelled base load (39–43 vs 16–27), consistent with fixed equipment loads (interpretation). Residences step down at 08:00 and up at 21:00. Use cases: hourly peak planning, and flagging buildings that exceed expected use.



8 · Limitations

Spikes beyond the labelled range are extrapolated linearly; that is supported by train spikes but is an assumption beyond them. Spikes in a *missing* feature cannot be recovered by imputation. No dates exist, so no external weather or calendar join was possible. Inference: numpy / pandas / LightGBM 4.5.0 from the base image, about 3 s.